100 Days of Linux - Day 032

Title: Removing Duplicate Lines with uniq

Today's Goal
Learn how to identify, remove, and count duplicate lines using the **uniq** command.

Why This Matters
Duplicate data is common in logs, reports, and datasets. Data engineers often combine **sort** and **uniq** to clean data.

Command of the Day: uniq

Syntax: uniq filename
Common Options:
uniq -c filename (count duplicates)
uniq -d filename (show only duplicate lines)
uniq -u filename (show only unique lines)

Important: uniq works best on **sorted** files.
Examples:
sort names.txt | uniq
sort names.txt | uniq -c

Beginner Lesson
The **uniq** command compares adjacent lines. Because of this, you should usually sort a file first so identical lines are grouped together.

Practice Questions

1. Create a file named names.txt with duplicate names. Sort it and remove duplicate lines.
2. Count how many times each name appears using uniq -c.
3. Display only duplicate names using uniq -d.
4. Display only names that appear exactly once using uniq -u.
5. Display your current working directory after completing today's tasks.

Hints
Always sort the file before using uniq to get accurate results.

Mini Challenge
Create employees.txt with repeated employee names. Use sort | uniq, sort | uniq -c, sort | uniq -d, and sort | uniq -u to analyze the file.

Common Mistakes
Running uniq on an unsorted file. Duplicate lines separated by other lines won't be detected.

Did You Know?
Many Linux data-cleaning pipelines begin with sort followed by uniq.

Pro Tip
The pipe symbol (|) sends the output of one command directly into another command.

Answer Key (Instructor Only)
Q1: sort names.txt | uniq
Q2: sort names.txt | uniq -c
Q3: sort names.txt | uniq -d
Q4: sort names.txt | uniq -u
Q5: pwd

Homework
Create three files with duplicate values and practice every uniq option until you understand the differences.

Next Day Preview
Tomorrow you'll learn the wc command to count lines, words, and characters in files.